



Rise of the machines: Delegating decisions to autonomous AI[☆]

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ARTICLE INFO

Keywords:

Decision delegation
Artificial intelligence
Social risk
Control premium

ABSTRACT

Delegation is an important part of organizational success and can be used to overcome personal shortcomings and draw upon the expertise and abilities of others. However, delegation comes with risks and uncertainties, as it entails a transfer of power and loss of control. Indeed, research has documented that people tend to under-delegate to other humans, often leading to poor decisions and ultimately negative economic consequences. Today, however, people are faced with a new delegation choice: Artificial Intelligence (AI). Fueled by Big Data, AI is rapidly becoming more intelligent and frequently outperforming human forecasters and decision-makers. Given this evolution of computational autonomy, researchers need to revisit the hows and whys of decision delegation and clarify not only whether people are willing to cede control to AI agents but also whether AI can reduce the under-delegation that is especially pronounced when people are faced with decisions that spur a high desire for control. By linking research on decision delegation, social risk, and control premium to the emerging field of trust in AI, we propose and find that people prefer to delegate decisions to AI as compared to human agents, especially when decisions entail losses (Studies 1–3). Results further illuminate the underlying psychological process involved (Study 1 and 2) and show that process transparency increases delegation to humans but not to AI (Study 3). These findings have important implications for research on trust in AI and the applicability of autonomous AI systems for managers and decision makers.

1. Introduction

Artificial Intelligence (AI) is reshaping the world and presents great opportunities for individuals and businesses. To date, most AI systems that are in wide use directly or indirectly operate under the responsibility of humans, who are in control of the analytical process or outcome. However, a growing number of AI systems go beyond acting as human proxies and operate in a truly autonomous manner. These systems are designed and empowered to make their own decisions fueled by the vast amounts of data they receive, analyze, and interpret (Service-Now, 2020). Such autonomous AI systems have the capability to surpass human intelligence across various industries and business functions, making them a powerful force for competitive advantage (Schrage, 2017). This technological progress creates entirely new opportunities for humans to delegate decisions to algorithms and artificial agents that no longer require human supervision or direction (Goldbach et al., 2019).

In business practice, Management by Algorithm (MBA) is becoming more commonplace, and many predict that the most successful

corporations will be those who delegate critical business decisions to smart algorithms (Schrage, 2017). Autonomous AI can determine entire marketing and capex strategies, identify competitors and target segments, personalize products and prices to customers, and customize communications to individualized preferences (Huang & Rust, 2021). For example, Renaissance Technologies, along with other investment funds, are relying on autonomous algorithms to analyze a situation, author a strategy, and execute it (Schrage, 2017). On an individual level, AI can automate bidding in online auctions (Adomavicius et al., 2009), trading in financial markets (Hendershott et al., 2011), and purchase decisions for customers, as well as automate and augment sales processes and frontline employee tasks (Grewal et al., 2020). Reliance on such new technologies can affect users' judgments and decisions, influence the magnitude of behavioral biases (Dowling et al., 2020; Herrmann et al., 2015), and thus substantially change and even improve decision making, business strategies (Davenport et al., 2020), and market outcomes (Herrmann et al., 2015). Given the vast applicability and the huge potential, some claim that organizations need to clarify when talented humans must defer to algorithmic judgment and delegate

[☆] This work was supported by the URPP Social Networks.

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decisions to (autonomous) AI (Schrage, 2017).

Although decision delegation encompasses many areas of people's private and professional lives, empirical evidence suggests that the decision to delegate and transfer power to others is not fully rational and often less than optimal. Indeed, investors often under-delegate to more skilled financial agents (Owens et al., 2014), or executives tend to micromanage subordinates, even if they have more specialized skills (Bartling et al., 2014). Often the reasons for such under-delegation boil down to concerns about the agents' knowledge, biases, or their intentions and self-interests (Bobadilla-Suarez et al., 2017; Butler & Miller, 2018), concerns which may be heightened for decisions that spur higher or lower desires to maintain control (Owens et al., 2014). The results are inefficient delegation choices, over-monitoring, and under-delegation that can lead to poor(er) decisions and thus give rise to harmful economic consequences.

Given the evolution of computational autonomy and the fact that AI systems increasingly outperform humans in a variety of tasks, it is important to understand how prior findings on decision delegation apply if the counterparty is an artificial rather than a human agent, or put differently, whether non-social agents change or even improve people's delegation choices. Yet, contemporary AI research has mainly focused on participative decision-making and the determinants of individuals' reliance on AI advice, such as its design (e.g., Dietvorst et al., 2018) or the context in which it is applied (e.g., Castelo et al., 2019). As AI can increasingly take on the role of an autonomous decision maker, research on delegation behavior, where individuals transfer their decision-making power and fully entrust AI to make decisions on their behalf (Goldbach et al., 2019), is largely missing. Additionally, while past and current AI research underlines the contextuality of people's trust in AI, it has largely limited its view to the task (e.g., subjective vs. objective, Castelo et al., 2019) or domain (e.g., medical services, Longoni et al., 2019). From the field of delegation research, however, we know that the choice to delegate largely hinges on the potential decision outcome itself. Thus, it is important to understand not only if users are more willing to cede control to AI than to human agents, but also why and how delegation varies for decisions that may spur a high or low desire to maintain control and thus to maintain decision making power.

The present research aims to close these gaps. In particular, we investigate whether people are (more) willing to delegate decisions to AI agents than to human agents. Moreover, we examine if peoples' choice to delegate a decision to a human or an AI agent differs for two different decision contexts: (1) decisions that spur a high desire for control, as they could lead to potential losses and, conversely, (2) decisions that could lead to gains but entail lower desires for control. In doing so, we are the first to link literature on decision delegation, social risk, and control premium to the increasingly important research on trust in AI.

Across three experiments, we provide empirical evidence for a general preference to delegate decisions to AI agents as compared to human agents in 'algorithm appropriate' domains, that is, domains that have a precise, external standard of accuracy (Logg et al., 2019). We show that this difference in delegation rates originates from a *social risk premium* (Butler & Miller, 2018) demanded of human delegates, especially when the potential decision outcome spurs a high desire for control. We show that the perception that humans, but not machines, are capable of intentional action gives rise to an uncertainty about the agent's intentions, which ultimately causes users to demand a higher expected payoff of delegation in order to cede control to human agents, as compared to AI agents. The findings of this research reveal the underlying psychological process leading to differences in delegation rates and provide potential remedies to mitigate delegation differences. In doing so, this research can offer insights for managers as to how AI agents can be employed strategically to improve (biases in) decision delegation and to avoid under-delegation, especially for circumstances and decisions that raise a desire for control.

In what follows, we first review prior work on trust in AI and decision delegation. We then develop a set of key hypotheses regarding how

decision delegation to AI and to human agents might differ. Next, we present the settings and results of three experiments designed to test our theorizing. Finally, we put our findings into context by deriving specific implications for theory and practice.

2. Review of the literature

Two lines of inquiry in the existing literature are of key interest to our own research. The first is the emerging field on trust in algorithms and AI. Based on their application and degree of autonomy, researchers categorize AI systems into two broad types (e.g. Komiak & Benbasat, 2006): advisory algorithms and performative AI. While advisory systems are used to *augment* human decision making, performative AI systems are used to *automate* decisions and actions without human input. Early research in the area has focused on advisory systems, also referred to as decision-support systems (Alter, 1977), interactive decision aids (Häubl & Trifts, 2000), and robo- or algorithmic advisors (Logg et al., 2019). As the names imply, these systems support and advise decision makers but ultimately leave the final decision up to the user. Much of the scholarly attention has been paid to *advice utilization*, or put differently, how strongly people adjust their own judgments following the advice of the system (Logg et al., 2019; Önkal et al., 2009). Initially, researchers uncovered a general aversion to reliance on statistical methods over human judgments (Dawes et al., 1989). As Logg et al. (2019) explain, "it appears that the experts in the 1950s were hesitant to believe that a linear model could outperform their judgment" (2019, p. 91). Yet decades later, studies show that especially lay people seem to put more weight on algorithmic over human advice when needing to make a decision (Logg et al., 2019).

Recently, research on performative AI has gained momentum, not least because of the rapid evolution of computational autonomy. Instead of supporting humans, these systems can autonomously perform a variety of tasks ranging from high-frequency trading to fraud detection, traffic planning, or pricing decisions (Shrestha et al., 2019). To date, people often seem averse to fully rely on such autonomous systems. In experiments by Dietvorst et al. (2015), for instance, participants preferred to exclusively use their own forecast rather than exclusively rely on an algorithmic forecast. Similarly, Logg et al. (2019) found that the *algorithm appreciation* documented in their studies waned when participants in an experiment had to choose between their own judgment and that of an algorithm. According to Bigman and Gray (2018), this aversion to delegate decisions to autonomous machines is especially pronounced for morally-relevant driving, legal, medical, and military decisions. A way to eliminate this aversion is to increase the machine's perceived experience and expertise (i.e., critical aspects of mind perception) and to limit the system to an advisory role (Bigman & Gray, 2018). In this vein, Dietvorst et al. (2018) suggest that the aversion to delegate to an imperfect algorithm can be overcome by *not* framing the decision to delegate "as an all-or-nothing" decision but rather by giving users a sense of control and ability to modify the forecast. While these studies seem to suggest that people are averse to delegate their full decision power to performative AI, it is important to note that people have always shown a tendency to trust more in their own intuitions as compared to others (Yaniv & Kleinberger, 2000) and to delegate decisions all too seldomly (Owens et al., 2014). It is hence unclear whether this aversion to fully delegate decisions to AI is driven by the fact that the advisor is an algorithm (i.e., *algorithm aversion*) or that people overly prefer to retain decision making power (i.e., *under-delegation*).¹

To understand what drives people's decision to delegate or not to delegate, we draw from our second line of inquiry in the literature: research on decision delegation. Decision delegation means entrusting

¹ Indeed, when comparing their findings across experiments, Dietvorst et al. note that "participants were more likely to bet on the model against another participant (Study 4) than against themselves (Study 1)" (2015, p. 8).

others to make a decision on one's behalf. Therefore, it is considerably different from participative decision making, as it entails a transfer of power from the individual who chooses to delegate (i.e., the *principal*, in the language of agency theory) to the one entrusted to make a decision on the former's behalf (i.e., the *agent*). As the principal essentially cedes decision-making authority to an agent, perceptions of the agent play a key role in decisions to delegate. In particular, research suggests that two basic requirements need to be met for delegation to occur: First, the agent needs to be sufficiently skilled, knowledgeable, and competent to make a good decision or effectively execute a task (Sengul et al., 2012); and second, the interests of the agent and the principal need to be well aligned, such that the agent does not act opportunistically or contrary to the principal's interest (Lupia, 2015). If these conditions are met, delegation can be used effectively to overcome limited time, energy, skills, and other personal shortcomings.

In theory, the decision to delegate should be fully rational, based on some form of cost-benefit analysis of these aforementioned factors. Yet, evidence is mounting that the decision to delegate is not always rational and is often less than optimal (Bobadilla-Suarez et al., 2017). In particular, research shows that delegation frequency is often too low. That is, people all too often accept lower expected payoffs and are overly reluctant to delegate decision rights (Dominguez-Martinez et al., 2014). One key explanation provided in the literature to explain under-delegation is the *control premium* (Owens, 2014). According to this proposal, people deliberately and consciously "pay" to maintain control and keep the decision right. For instance, in an experimental task where participants could bet that a partner or they themselves will answer questions correctly, people bet on themselves 64.9% of the time, whereas it turns out they should have bet on themselves only 56.4% of the time (Owens et al., 2014). This preference for control does not simply reflect overconfidence or subjective beliefs (Owens, 2014). Instead, it seems to be mainly driven by people's loss aversion and fear that something may go wrong combined with an (irrational) fear that agents may misbehave (Dominguez-Martinez et al., 2014).

More recently, another explanation for under-delegation has been proposed: the *social risk premium* (Butler & Miller, 2018). This research suggests that alongside an aversion to cede control, people also have an aversion to the risk of betrayal, thus leading people to demand a premium before they are willing to expose themselves to a social risk. Most generally, this 'social risk' refers to instances where decisions by other human beings are the prime source of uncertainty (Bohnet et al., 2008). Clearly, such social risks also arise in delegation decisions, as the individuals must decide whether to trust the agent and make themselves vulnerable to that other person. Hereby, the counterparty's intentions play a substantial role in determining principals' attitudes towards exposing themselves to this social risk (Butler & Miller, 2018). In fact, Butler and Miller (2018) show that the *social risk premium* people demand to delegate a decision and thus expose themselves to the social risk of another person does not arise from the mere fact that a human being is the source of risk. Instead, the social risk premium is driven more specifically by an aversion to a counterparty's potentially malign intentions (i.e., an instance of the more general aversion to cede control to someone with conflicting interests) and relates to a counterparty's capacity to engage in intentional action. To put it in Butler and Miller's words, "the intentional capacity of the counterparty is crucial" (2018, p. 2793).

Most work on decision delegation has focused on managers delegating decisions or tasks to subordinate employees (Steffel et al., 2016). While a number of these studies have examined the impact of agent characteristics on the decision to delegate, the focus has predominantly been on various attributes of human agents (e.g., gender: Akinola et al., 2018; age: Sengul et al., 2012). Research examining the effect of different types of agents (in particular, AI vs. human agents) is almost non-existent in this literature (Belanche et al., 2020). In the following, we will discuss how the type of agent (AI vs. human) may affect the control and social risk premiums and thus, ultimately, under-delegation.

3. Hypotheses

The present research suggests that autonomous AI can reduce under-delegation in two important ways: First, AI agents alleviate the social risk premium and people's fear of betrayal, as these agents are perceived to be incapable of intentional action. Second, people's preference for control (i.e., the control premium) will amplify delegation differences between AI and human agents because people *feel* more in control when delegating to a machine. Fig. 1 illustrates our theoretical framework. We will detail our reasoning in the sections below.

3.1. The effect of AI on the social risk premium

As noted above, individuals generally exhibit an aversion to expose themselves to a social risk and thus willingly accept lower levels of payoff rather than delegate decisions to a human counterpart. Thus, when faced with the same payoffs and probabilities, individuals are generally less willing to delegate when the outcome depends on whether another person proves trustworthy than when the outcome is due to a chance device (i.e., posing a natural risk) (Bohnet et al., 2008). One key aspect giving rise to this social risk premium is the intentionality of humans (Butler & Miller, 2018).

Contemporary AI research suggests that people do not tend to associate algorithms, artificial agents, or AI systems with autonomous goals and the capacity for intentional actions. Indeed, the seminal work by Gray et al. (2007) on mind perception and mental capacities demonstrates that people believe robots are incapable of experience in general and incapable of experiencing fear, hunger, desire, pride, or embarrassment in particular. People also believe that robots have considerably lower levels of agency than humans, which includes capacities such as thought, planning, or self-control. In fact, robots were ascribed similar levels of experience and agency as that ascribed to a dead person (Gray et al., 2007). Unsurprisingly, more recent research found that people believe that AI agents do not have self-driven goals (Kim & Duhachek, 2018), have less intentional capacity (Harbers et al., 2017), and, in consequence, are less self-interested and more unbiased (Cardinaels et al., 2017; Fogg, 2009).

Linking the notion of social risk to the low intentional capacity ascribed to AI agents suggests that these agents should pose a lower social risk in the context of decision delegation, as uncertainties about the agent's maligned intentions are unlikely to come to mind. For this reason, the social risk premium, which people demand in order to cede control to an agent with high intentional capacity, should be lower for AI agents. As a result of this reduced social risk premium, we expect that individuals will delegate more decisions to AI agents than to human agents. In short, we hypothesize:

H1. Delegation rates are higher when the decision can be delegated to an AI system rather than to a human.

H2. The effect of the agent type on rates of delegation is mediated by the perceived intentional capacity of the agent.

3.2. The effect of AI on the control premium

In addition to the social risk premium, a control premium further aggravates people's propensity to under-delegate. According to Owens et al. (2014), the control premium reflects a preference for control which inhibits individuals' willingness to delegate decisions that affect their lives, even if this preference reduces the payoffs or has disadvantageous consequences for organizational efficiency. While many motivations may underlie people's preference to maintain decision-making control (e.g., intrinsic utility of authority: Bartling et al., 2014), recent research suggests that the control premium is largely driven by individuals' loss aversion and the fear that something may go wrong when delegating a decision (Dominguez-Martinez et al., 2014). This desire for control is hence influenced by outside events and circumstances (Ramsey &

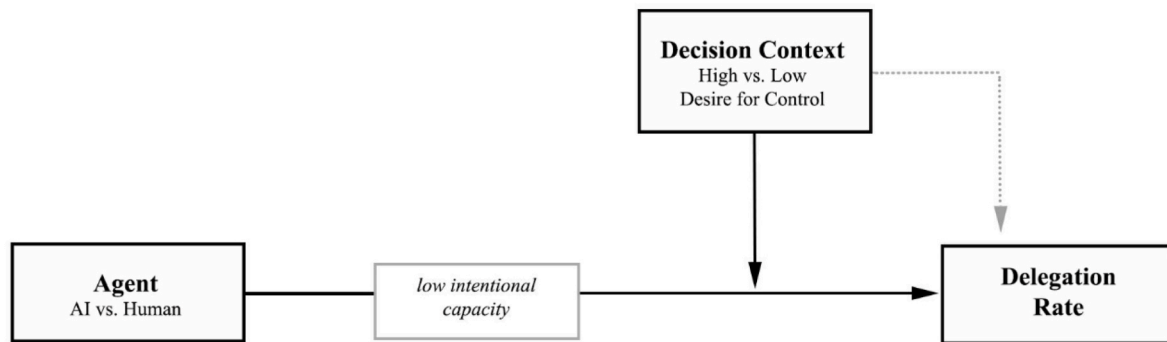


Fig. 1. Conceptual Framework of this Research. Relationship between agent type, intentional capacity, desire for control, and delegation rate.

Etcheverry, 2013).

The decision context, or more precisely the potential decision outcome, is one such circumstantial variable that influences people's desire for control. A person might need to decide between two aversive outcomes, such as selling a stock or holding on to it in order to minimize a loss, or between two rewarding outcomes, such as selling a stock or holding on to it in order to maximize a gain. Prior research suggests that especially losses spur an excess desire for control in this way, ultimately leading to irrational delegation choices and under-delegation (Dominguez-Martinez et al., 2014; Ramsey & Etcheverry, 2013).

It is important to note, however, that there is a difference between objectively being in control and subjectively feeling in control. Indeed, research suggests that the threat of aversive events can be mitigated by a perception of being in control (Friedland et al., 1992). Thus, instead of actually being or staying in control, the subjective perception of 'feeling in control,' or more precisely the feeling of not losing control, is what ultimately determines individuals' decision to delegate (Spreitzer & Mishra, 1999). According to Spreitzer and Mishra (1999), managers' trust that delegates will not act opportunistically as well as consideration of delegates' prior work performance can reduce the uncertainty about the outcomes to be expected from delegation and can, hence, serve as alternatives to formal control. In short, if people feel they are still in control or believe they will not lose control, they are more likely to delegate a decision.

Interestingly, contemporary research on AI and algorithmic advice suggests that people consider the performance of technological systems to be more stable and thus less unpredictable than human action (Belanche et al., 2020). This suggests that people experience less uncertainty about the outcomes to be expected from delegation. Moreover, prior research suggests that individuals perceive themselves to maintain their autonomy, and thus to be more in control, when interacting with AI instead of humans. Research on automated helpers, for instance, has demonstrated that people are more likely to accept advice from non-humanized agents than humanized agents, as the former lack social agency and, therefore, are not perceived to undermine a person's autonomy (Kim et al., 2016). Similarly, research in robotics has shown that reduced social cues in persuasive social robots leads to lower perceived threat to autonomy and, in consequence, greater compliance (Ghazali et al., 2018; Roubroeks et al., 2011). This is in line with the suggestion that technology products are typically perceived as tools designed to help accomplish a task (Hinds et al., 2004).

Taken together with the above, it follows that the lower social agency of AI agents as opposed to human agents makes delegating decisions to AI agents feel more predictable and controllable. As people feel they are not losing as much control when delegating a decision to AI as opposed to another human, they should also be more likely to delegate decisions in this way even when the situation increases the individuals' desire for control. Following this line of thought, we propose that the effect of agent type on delegation rates will be amplified when people are faced with delegation situations that spur a high desire for

control: When decisions can only be delegated to a human, decisions characterized by a high desire for control will foster under-delegation. In contrast, when decisions can be delegated to AI (perceived to be low in agency, thus leaving individuals' sense of autonomy and control intact), situations that spur an excess need for control will not have such a detrimental effect on delegation rates. In short, we hypothesize:

H3. The differences in delegation rates between AI and human agents will be larger for decisions that could result in a loss (i.e., spurring a high desire for control) and smaller for decisions that could result in a gain (i.e., spurring a low desire for control).

H3a. When decisions can be delegated to a human, delegation rates will be lower for decisions entailing a loss (i.e., spurring a high desire for control) and higher for decisions entailing a gain (i.e., spurring a low desire for control).

H3b. When decisions can be delegated to an AI system, delegation rates will not differ between decisions entailing a loss (i.e., spurring a high desire for control) and decisions entailing a gain (i.e., spurring a low desire for control).

4. Study 1

The aim of Study 1 is to examine potential differences in delegation rates to human versus AI agents and how these delegation rates change when the decision outcome poses a potential loss rather than a potential gain. Furthermore, we examine the perceived intentional capacity of the different agents to gain a better understanding of why differences in delegation rates may arise.

4.1. Design and procedure

Study 1 used a 2×2 (AI vs. human; gain vs. loss) between-subjects experimental design. A total of 500 people were invited to the study. We excluded participants from data analyses if 1) they failed to pass a simple attention check at the beginning of the study, or 2) they were identified as speeders using Mahalanobis Distance (with threshold at 0.999 quantile of the χ^2 p). On these grounds, 8 participants (1.6% of the sample) were excluded, resulting in a final sample of 492 participants (US MTurkers, $M_{age} = 40$ years, 38% females). Participants were randomly assigned to a loss or gain setting and to a human or AI agent. Across all conditions, participants had to complete two parts: Part I was a learning task, where participants were first familiarized with the task, and Part II was a delegation task, which we used to measure our central dependent variable, the delegation rate. The behavioral design of the experiment was adapted from prior research by Bobadilla-Suarez et al. (2017) and aimed to capture real delegation behavior rather than mere delegation intentions. Fig. 2 provides a graphic representation of the task.

Part I of the study consisted of 16 trials. In each trial, participants viewed for 1 s a square that was filled with red dots and a separating

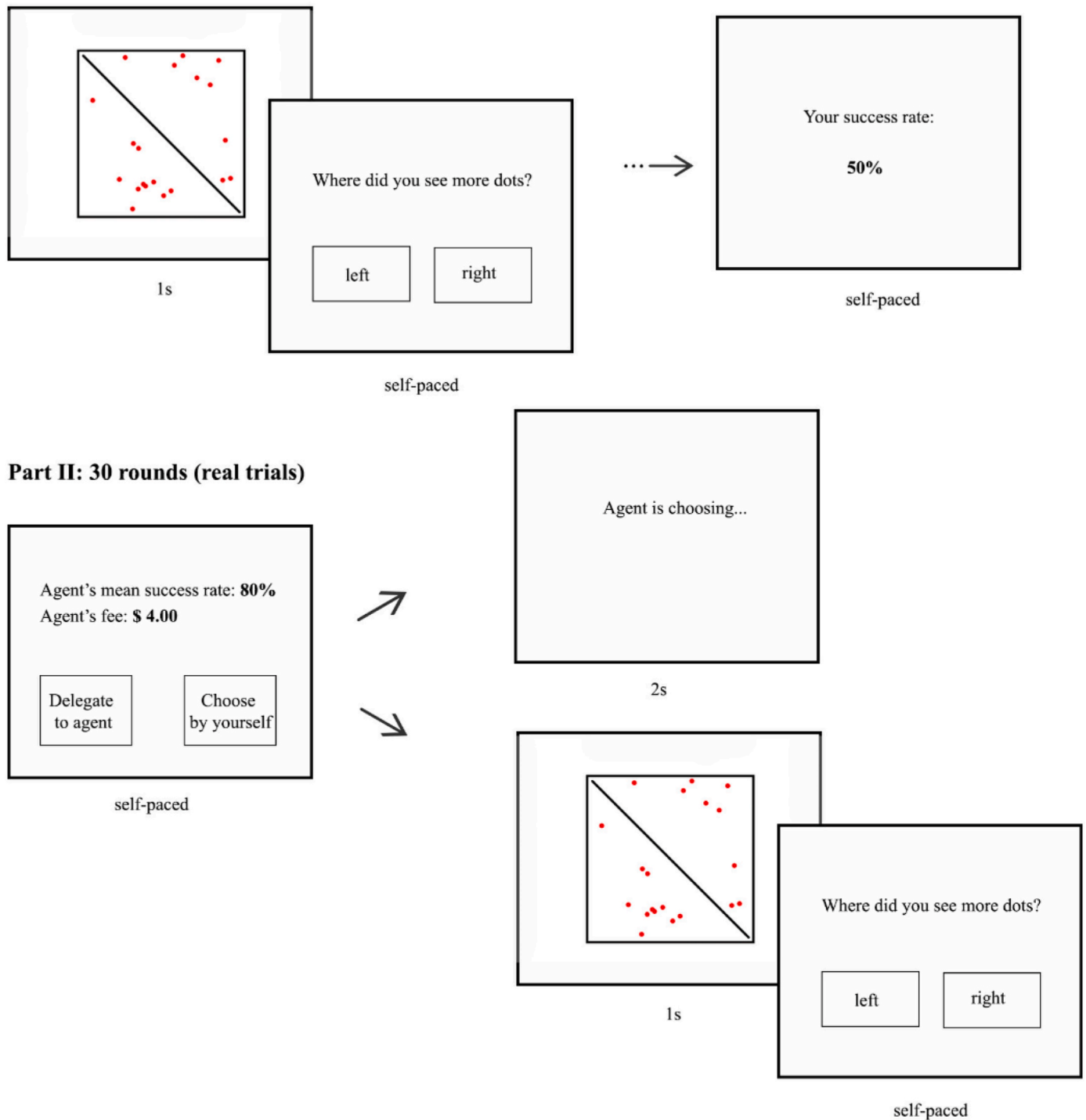


Fig. 2. Graphic representation of the task. In the first part of the experiment, participants were asked on each trial to choose between two sides. After 16 trial rounds, they were told that they chose the correct side in 50% of the rounds. In the second part of the experiment, participants first chose whether they would like to make the choice by themselves or delegate the task to an advisor. They were presented with the advisor's success rate and the fee. Outcomes were not revealed.

diagonal line. This task is closely related to the figure task used to study delegation decisions to algorithms by Bobadilla-Suarez et al. (2017). Participants had to estimate if there were more dots to the left or the right of the separating line. We considered 1 s enough time for participants to gain a sense of the dots' distribution but not enough time for participants to count the dots on each side. The dots were distributed differently each time and in a way that a correct answer was not easily apparent to the viewer, thus making delegation of the task a viable option. After the trial rounds, all participants were told that their own

success rate was 50%, meaning that they chose the correct side in 8 out of 16 trials. In Part I of the study, the success rate did not affect payment to participants.

In Part II of the study, we randomly assigned participants the ability to delegate the previously introduced square task either to an AI agent (for half of the participants) or to a human agent (the other half of participants). The goal of Part II was to quantify participants' willingness to delegate decisions to an agent for the same task they had undertaken in Part I. The exact wording supplied in the human vs. AI agent

conditions is provided in Table 1. As perceived (in)competence of the agent can largely affect delegation rates, we also provided information on the agents' performance rate. To avoid any potential confounds, the performance rates did not differ between AI and human conditions. Given that we also provided information about the participants' own performance, misperceptions of their own abilities compared to the abilities of the potential agent would not have influenced the actual delegation decision.

Part II consisted of 30 trials. For each trial, participants first had to decide whether to make the decision by themselves or to delegate the decision to an agent. To avoid learning effects and to make sure that every delegation choice was made irrespectively of the previous decision, we followed prior research and let participants delegate to a different agent (although only of the same type) on every trial (Bobadilla-Suarez et al., 2017). For each trial, participants were presented with two pieces of information about the agent: 1) the agent's mean success rate in the task to avoid uncertainties or different perceptions about the agent's performance (kept constant across all rounds at 80% for both AI and human agents) and 2) the agent's fee (from \$0 to \$6). If participants decided to choose for themselves instead of relying on an agent, they were shown the square task for a second and then given unlimited time to make their decision (as in Part I of the study). If a participant decided to delegate, a new screen appeared for 2 s saying, "Agent is choosing ..."

We also randomly assigned participants to either a loss or gain condition for these 30 trials. The payments involved were independent of the base payment that all participants in the study received. In the gain condition, a correct choice would result in a monetary gain and an incorrect choice in no gain. In the loss condition, a correct choice would lead to no loss and an incorrect choice in a monetary loss. Furthermore, in the gain condition, participants started with a "bonus" budget of \$0, and they could gain extra payments in each of the 30 trials. In each round, when a correct choice was made, the participant in the gain condition could gain up to \$10 (desirable outcome); when the wrong side was selected, the participant earned nothing extra (i.e., they received \$0). In the loss condition, participants started with a "bonus" budget of \$300 and could lose extra payments in each of the 30 trials. The desirable outcome (when the correct side was chosen) was \$0 and the undesirable outcome (when the wrong side was chosen) was -\$10 (Bobadilla-Suarez et al., 2017).

As we had told participants that their own success rate was 50% in Part I of the study, their expected value of choosing by themselves was \$5 in the gain condition ($50\% \times (\$10)$) and -\$5 in the loss condition ($50\% \times (-\$10)$). In each trial, an agent's fee ranged from \$0 to \$6 and was presented to the participants in random order. All agents had the expected success rate of 80%. The participants paid the fee only when they delegated the decision to the agent and the agent selected the correct side. Hence, the expected value of using an agent is calculated as the agent's *success rate* * *desirable outcome* minus the *agent's fee*. For example, in the gain condition an advisor with 80% accuracy and a fee of \$5 would return an expected value of $((90\% \times \$10) - \$5) = \$4$. In this case, a value maximizer would not delegate the decision to the advisor. The

expected value of using an agent ranged from \$2 (if the agent's fee was \$6) to \$8 (if the agent's fee was \$0), with a mean of \$5 ($SD = 1.65$) in the gain condition, and from -\$2 (if the agent's fee was \$0) to -\$8 (if the agent's fee was \$6), with a mean of -\$5 ($SD = 1.65$) in the loss condition.

To increase participants' stakes in these tasks, we told participants that we paid out any extra earnings in these 30 trials as bonus payments (divided by 100) after the study, meaning participants in the gain condition received everything they earned and participants in the loss condition received everything they did not lose. We followed the procedure done by prior research on valence dependent delegation (Bobadilla-Suarez et al., 2017).

4.2. Measurements

We calculated the delegation rate (i.e., the dependent variable) as the percentage of trials in which participants chose to delegate (Bobadilla-Suarez et al., 2017). Additionally, after the delegation task and before they learned about their final payout, we measured the agent's intentional capacity on a 7-point Likert scale with two items adapted from Williams et al. (2004), which capture the participant's perception of the agent's intentions and self-interests ("The advisor has self-interested profit motives" and "The advisor acts in his own best interest," where 1 = strongly disagree and 7 = strongly agree; and a semantic differential scale "The advisor acts in (1) his own best interest to (7) my best interest") ($\alpha = 0.74$).

4.3. Manipulation check

Social agency indicates the degree to which a social agent is perceived as a social entity, meaning the degree to which an agent is perceived as being capable of social behavior that resembles human-human interactions. In order to measure how much social agency the participants attributed to the agent, we used two statements measuring the capacities of agency on a 7-point Likert scale ranging from 1 (fully disagree) to 7 (fully agree) (adopted from Gray et al., 2007). The statements were "The advisor is capable of making plans and working toward a goal" and "The advisor is capable of experiencing emotions" ($\alpha = 0.77$). Results showed that our agent manipulation was successful and that participants attributed more social agency to the human ($M = 6.21$) than to the AI agent ($M = 3.15$), $t(439.57) = -24.267$, $p < .001$.

4.4. Results

Analysis of delegation rates. To test H1, we first examined the main effect of agent type on delegation rates. Results of an independent *t*-test with agent type as the independent variable and delegation rate as the dependent variable showed that participants in the AI agent condition displayed significantly higher delegation rates compared to participants in the human agent condition ($M_{AI} = 0.47$, $M_{human} = 0.35$, $t(489.07) = 4.41$, $p < .001$). This is in line with H1, where we proposed that delegation rates would be higher for AI agents as compared to human agents.

Further, an independent *t*-test with agent type as the independent and self-interest as the dependent variable showed that the human agent was perceived to act more from self-interest ($M = 4.86$) than was the AI agent ($M = 3.03$), $t(483.06) = -13.28$, $p < .001$.

A simple mediation following the procedure of Preacher et al. (2007) was used to analyze H2. In the model, delegation rate was the dependent variable, agent type (0 = human, 1 = AI) the independent variable, and self-interest the mediator. The path (direct effect) from agent type to self-interest was negative and statistically significant ($\beta = -1.835$, $p < .001$). The path (direct effect) from agent type to delegation rate was positive and significant ($\beta = 0.066$, $p = .036$), indicating that AI agents increase delegation rate. The direct effect of self-interest on delegation rate was negative and significant ($\beta = -0.029$, $p = .001$), indicating that delegation rate is reduced when an agent is perceived to be more self-interested. We used Preacher and Hayes's (2008) bootstrapping

Table 1
Description of agent for human vs AI agent.

Condition	Wording
Human agent	"Human advisors are humans trained for the task. Their decisions are based on their personal expertise. On each trial there will be a different advisor and the advice they offer is drawn from the responses that have already been computed previously."
AI agent	"AI advisors are machines trained for the task. Their decisions are based on mathematical rules and algorithms. Predictions are made by analyzing large data sets with algorithms and the process involves no human inference. On each trial there will be a different AI algorithm and the advice they offer is drawn from the responses that have already been computed previously."

procedure to obtain unbiased 95% confidence intervals around the mediating effect of self-interest. Confidence interval (CI) for the indirect effect excluded zero (indirect effect = 0.054, CI = [0.020 to 0.087]). This shows that self-interest significantly mediated the relationship between agent type and delegation rate, a result which supports H2.

In H3 we hypothesized that the control premium affects delegation rates for human agents but not so for AI agents. To test for the moderating effect of the potential decision outcome (loss vs. gain), we used a 2×2 full factorial ANOVA with delegation rate as dependent variable and agent type (AI vs. human) and outcome domain (gain vs. loss) as the dependent variables. Results revealed a significant main effect for agent type ($F(1,488) = 19.62, p < .001$), an insignificant main effect for outcome domain ($F(1,488) = 1.45, p = .230$) and a significant interaction effect of agent and outcome domain ($F(1,488) = 4.76, p = .030$). Analyzing the interaction effect showed that in the loss condition, delegation rates for AI agents were significantly higher ($M = 0.48$) than for human agents ($M = 0.31$), $t(251.86) = 4.93, p < .001$. Delegation rates in the gain condition were in the same direction; that is, delegation rates for AI agents were higher ($M = 0.46$) than for human agents ($M = 0.40$), but this difference was not large enough to be statistically significant ($t(229.57) = 1.46, p = .144$). Moreover, in support of H3a, we found that for human agents, delegation rates were significantly higher in the gain condition ($M = 0.40$) than in the loss condition ($M = 0.31$), showing a domain-dependent effect on delegation rates ($t(222.25) = 2.41, p = .017$). For AI agents, delegation rates did not differ between the gain ($M = 0.46$) and the loss ($M = 0.48$) conditions, ($t(249.63) = -0.66, p = .507$). This is in line with H3b.

As our studies recorded age and gender of our participants, we checked if there is any relationship between these factors and our obtained results. Results of an ANOVA model including the two factors shows that age ($F(1,485) = 0.78, p = .377$) and gender ($F(1,485) = 0.44, p = .642$) have no effect on our findings.

Mixed-model analysis of delegation probability. Following previous research on choice delegation, we also ran a mixed-effects model to account for additional information in our data (Bobadilla-Suarez et al., 2017).² Our outcome variable was the probability that a participant would delegate (delegation = 1) measured at 30 different values of the agent's expected value. Every participant had his/her own parameter estimate for the agent's expected value (modeled as a random slope) and for an intercept (also modeled as a random effect). We controlled for trial number to account for linear temporal effects such as disengagement over time. Random effects were also included for trial number.

Results of the mixed-effects model provide further support for our hypotheses. Thus, the results show a positive main effect of agent's expected value on delegation probability, meaning that an increase in agent's expected value leads to an increase in delegation probability ($\beta = 2.15, SE = 0.09, z(14760) = 23.30, p < .001$). Moreover, the interaction between agent type and outcome domain was significant ($\beta = -2.32, SE = 0.94, z(14760) = -2.46, p = .014$) (see Table 2). To analyze the interaction effect, we ran two separate models, one for AI agents and one for human agents. For human agents, outcome domain significantly predicted delegation probability: Delegation probabilities were significantly lower in the loss condition and higher in the gain condition ($\beta = -1.79, SE = 0.77, z(7200) = -2.34, p = .019$) (cf. H3a). For AI agents, outcome domain had no statistically significant effect on delegation probability ($\beta = 0.52, SE = 0.60, z(7560) = 0.87, p = .383$) (H3b) (see

Table 2

Results of Study 1: Full mixed-effects model of delegation. Estimates of the fixed effects, with standard errors and significance. Fixed effects are agent's expected value (success rate*desirable outcome – fee), agent type (0 = AI, 1 = human), outcome domain (0 = gain, 1 = loss), and trial number.

Mixed-Effects Model of Delegation Agent	
	Dependent variable: delegation
Agent's value	2.150*** (0.092)
Agent (Human)	-1.004 (0.674)
Domain (Loss)	0.580 (0.645)
Trial number	-0.045*** (0.011)
Agent:Domain	-2.316* (0.941)
Constant	-10.505*** (0.683)
Observations	14,760
Log Likelihood	-3971.513
Akaike Inf. Crit.	7963.026
Bayesian Inf. Crit.	8039.023

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 3

Results of Study 1: Separate mixed-effects models of delegation for AI and human agent. Separate estimates of the fixed effects, with standard errors and significance for the mixed effects model in the human and the AI condition. Fixed effects are agent's expected value (success rate*desirable outcome – fee), outcome domain (0 = gain, 1 = loss), and trial number.

Mixed-Effects Models of Delegation for Human and AI Agent		
	Dependent variable: delegation	
	Human (1)	AI (2)
Agent's value	2.294*** (0.152)	2.050*** (0.119)
Domain (Loss)	-1.793* (0.766)	0.523 (0.599)
Trial number	-0.049*** (0.011)	-0.040** (0.013)
Constant	-12.230*** (0.992)	-10.068*** (0.776)
Observations	7200	7560
Log Likelihood	-1785.803	-2191.011
Akaike Inf. Crit.	3587.606	4398.022
Bayesian Inf. Crit.	3642.661	4453.467

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 3).

4.5. Discussion

The findings of our Study 1 demonstrate that people believe that AI agents, but not human agents, lack self-interested motives. This finding is in line with contemporary AI research that suggests AI is perceived to lack self-interest and to be more unbiased (Cardinaels et al., 2017; Fogg, 2009). Moreover, the results of Study 1 show that participants are more willing to delegate decisions to AI, whereas they are less inclined to expose themselves to the social risk of human agents. Thus, delegation rates for human agents are lower, especially in the domain of losses, as the potential risk of an agent's betrayal looms larger (betrayal aversion). For AI agents, we do not find that this betrayal aversion affects delegation rates, which is in line with our theorizing that AI agents are not believed to act self-interested or have malign intentions. Given their importance, we will examine perceptions of risk and control more closely in the subsequent study.

5. Study 2

The objective of Study 2 is to achieve a better understanding of the underlying process leading to the differences in delegation rates

² ANOVA and MANOVA analyses for repeated measures research designs are nomothetic approaches that focus on group means. In contrast, newer multi-level modeling techniques are more informative than ANOVA and MANOVA, as they characterize both group-level (nomothetic) and individual-level (idiographic) effects. Hence, they yield a more complete understanding of the phenomena under investigation (Kristjansson et al., 2007). Mixed logit models are designed for binomially distributed outcomes and repeated measures designs (Jaeger, 2008).

between human and AI agents. In particular, Study 2 investigates participants' desire for control, which we suggested is heightened for decision outcomes leading to potential losses. In H3 we proposed that decisions that spur a high desire for control would increase the differing effect of agent type on delegation rates. We further suggested that this effect is driven by the fact that delegation to AI agents is perceived to be more controllable and less risky. We have, therefore, included measures of these perceptions in Study 2 to verify our underlying reasoning. In addition, Study 2 examines the mediating role of intentional capacity and the moderating role of the decision outcomes on delegation rates, using a moderated mediation model.

5.1. Design and procedure

Study 2 involved a total of 550 participants 15 of which were removed from the sample following the same rules as in the previous study (2.73% of the sample), resulting in a final sample of 535 participants (US MTurkers, $M_{Age} = 37$ years, 38% females). Participants were instructed to carry out the same decision tasks as in Study 1. The procedure of Study 2 thus largely follows that of Study 1, with the exception of a number of additional measures following the delegation task and before the participants' learned about their final payout: First, we included an adjusted one-item measure of desire for control that captured how important participants deemed it to be that they have control over the decision (1 = not at all important, 7 = very important) (adopted from Shapiro, 1994). Second, we added a measure of expected locus of control (a three-item scale modified from Dabholkar, 1994) to capture how much participants would perceive to be in control when delegating their decision to an AI vs. human agent. In particular, the items asked: "When you decide to delegate a decision to a human agent, who do you think is more in control of the process," "When you delegated tasks, who had more control over the outcome of the task" (1 = advisor, 7 = you), and "When you decide to delegate a decision to a human/AI advisor, how much control will you have over the outcome?" (1 = very little control, 7 = a lot of control) ($\alpha = 0.78$). Third, we adapted an item from Gürhan-Canli and Batra (2004) to measure the degree to which a person believes that the delegation to the respective agent involves a high risk ("Delegating a decision to a human advisor involves a lot of risk," 7-point Likert scale where 1 = completely disagree, 7 = completely agree). Finally, the self-interest of the agent was measured with the same items as in Study 1 ($\alpha = 0.83$) (scales adapted from Williams et al., 2004).

5.2. Manipulation check

The perceived social agency of the agent was measured with the same scale as in study 1 ($\alpha = 0.77$) (Gray et al., 2007). Results showed that manipulation was successful and that participants attributed more social agency to the human ($M = 6.03$) than to the AI agent ($M = 3.25$), ($t(478.54) = -21.658$, $p < .001$).

5.3. Results

Analysis of delegation rates. We first examined the main effect of agent type on delegation rates via an independent t -test with agent type as the independent variable and delegation rate as the dependent variable. Delegation rates were significantly higher in the AI agent condition ($M = 0.49$) than in the human agent condition ($M = 0.43$), $t(532.77) = 2.30$, $p = .022$ (H1).

We used two independent t -tests with agent type as the independent variable and perceived risk and expected control as dependent variables. As expected, participants perceived that delegating to a human agent involved higher risk ($M = 4.63$) than delegating to an AI agent ($M = 4.13$, $t(526.11) = -3.823$, $p < .001$), while expected control over the process was higher when delegating to AI agents ($M = 3.42$) than to human agents ($M = 3.07$, $t(531.59) = 2.283$, $p = .023$). Furthermore,

results of an independent t -test with outcome domain as the independent variable and desire for control as the dependent variable showed that desire for control was higher in the loss condition ($M = 5.38$) as compared to the gain condition ($M = 4.97$, $t(509.69) = -3.035$, $p = .003$).

To investigate the mediating role of intentional capacity on delegation rates and the moderating role of decision outcomes on the relationship of self-interest and delegation rates, we conducted a moderation analysis following Preacher et al. (2007). The outcome variable for analysis was delegation rate, the predictor variable was agent type (0 = human, 1 = AI), the mediator variable was perceived self-interest of the advisor, and the moderator variable was decision outcomes (0 = gain, 1 = loss). The results of the analysis showed that agent type had a significant effect on perceived self-interest ($\beta = -1.532$, $p < .001$), in accordance with H2. That is, perceived self-interest was significantly higher for human agents ($M = 4.83$) than for AI agents ($M = 3.30$, $t(525.04) = -11.504$, $p < .001$). The direct effect of agent type on delegation rate was insignificant ($\beta = 0.009$, $p = .765$). The full model further showed that decision outcome moderate the relationship of self-interest and delegation rate ($\beta = -0.036$, $p = .015$). A bias-corrected bootstrap analysis with 10,000 samples confirmed a full mediation model, as the 95% confidence interval (CI) for the indirect effect excluded zero (indirect effect = 0.055, $CI = [0.012$ to $0.101]$). The conditional indirect effects of self-interest on delegation rates showed that moderation was significant in the loss condition ($\beta = -0.050$, $CI = [-0.071$ to $-0.028]$, $p < .001$) but insignificant in the gain condition ($\beta = -0.014$, $CI = [-0.036$ to $0.009]$, $p = .228$).

To further examine the interaction effect, we graph the interaction effect in Fig. 3. As can be seen, in the loss condition, perceived self-interest has a strong negative effect on delegation rate. This negative relationship is weaker in the gain condition. Finally, delegation rates for human agents were significantly lower for decisions entailing a loss ($M = 0.39$) instead of a gain ($M = 0.48$, $t(256.36) = 2.496$, $p = .013$), in accordance with H3a. For AI agents, delegation rates did not differ for the loss ($M = 0.50$) and gain ($M = 0.48$) conditions $t(270.95) = -0.505$, $p = .545$, in accordance with H3b.

To test the interaction between potential decision outcome and agent type, we used the same 2×2 full factorial ANOVA as in Study 1, with

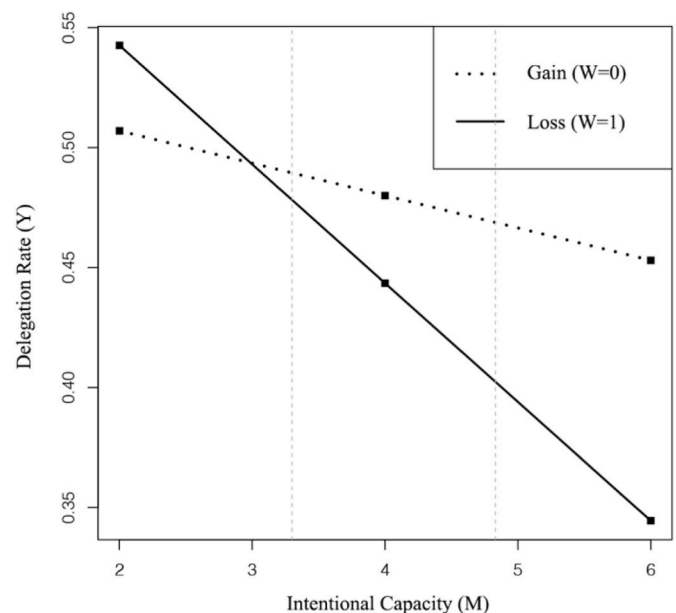


Fig. 3. A visual representation of the moderation of the effect of the agent's perceived self-interest on delegation rate by decision outcome. The gray lines show the average value of perceived self-interest for AI agents ($M = 3.30$) and human agents ($M = 4.83$).

delegation rate as dependent variable and agent type (AI vs. human) and outcome domain (gain vs. loss) as the dependent variables. This test showed a significant main effect for agent type ($F(1,531) = 5.33, p = .021$), an insignificant main effect for outcome domain ($F(1,531) = 1.53, p = .216$), and a significant interaction effect of agent and outcome domain ($F(1,531) = 4.64, p = .032$), replicating the findings of Study 1. In the loss condition, delegation rates were significantly higher for AI agents ($M = 0.50$) than for human agents ($M = 0.39$), $t(262.45) = 3.25, p = .001$. In the gain condition this difference was not significant ($M_{AI} = 0.48, M_{human} = 0.48$; $t(266.79) = 0.07, p = .946$). Moreover, for human agents delegation rates were domain dependent, thus delegation rates were significantly higher in the gain condition ($M = 0.48$) than in the loss condition ($M = 0.39$), $t(256.36) = 2.50, p = .013$ (cf. H3a). For AI agents, this difference was not significant ($M_{Gain} = 0.48, M_{Loss} = 0.50$; $t(270.95) = -0.61, p = .545$) (cf. H3b).

As in study 1, we ran our ANOVA model again, this time including age and gender. The effect of age ($F(1,529) = 0.15, p = .697$) and gender ($F(1,529) = 1.15, p = .285$) were insignificant and did not change our findings.

Mixed-model analysis of delegation probability. To analyze subjects' delegation probabilities, we used the same mixed-effects model as in Study 1. The full model showed a main effect for agent type and an interaction effect of agent type and outcome domain (see Table 4). To further analyze the interaction effect, we ran the analysis separately for AI agents and human agents. The first model showed that for human agents, the potential decision outcome (gains vs. losses) significantly predicted delegation probabilities. Delegation rates were lower in the loss condition ($M = 0.39$) than in the gain condition ($M = 0.48$) (cf. H3a). For AI agents, the decision outcome had no significant effect on delegation rates ($M_{loss} = 0.50, M_{gain} = 0.48$) (cf. H3b) (see Table 5).

5.4. Discussion

The results of Study 2 confirm the underlying theoretical assumption that desire for control is higher when facing potential losses. The study's findings further demonstrate that AI agents are not only perceived to have lower intentional capacity, but also that delegation to an AI agent is perceived to involve less risk and be more controllable than delegation to a human agent. Furthermore, we showed that the intentional capacity of the advisor fully mediates the relationship between agent type and delegation rate, and the effect of intentional capacity on delegation rate is moderated by outcome domain. The intentional capacity of the advisor thus has a stronger negative effect on delegation rates in the domain of losses, as compared to the domain of gains. Ultimately then, the difference in delegation rates for AI versus human advisors is greater for decisions entailing potential losses than those entailing gains.

Table 4

Results of Study 2: Full mixed-effects model of delegation. Coefficients, standard errors, and significance for the mixed effects model. The fixed effects are agent's expected value, agent type (0 = AI, 1 = human), outcome domain (0 = gain, 1 = loss), and trial number.

Mixed-Effects Model of Delegation	
	Dependent variable: delegation
Agent's value	2.143*** (0.087)
Agent (Human)	-0.342 (0.583)
Domain (Loss)	0.440 (0.571)
Trial number	-0.033*** (0.008)
Agent:Domain	-1.837* (0.818)
Constant	-10.077*** (0.601)
Observations	16,050
Log Likelihood	-4667.541
Akaike Inf. Crit.	9355.083
Bayesian Inf. Crit.	9431.918

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 5

Results of Study 2: Separate mixed-effects models of delegation for AI and human agent. Coefficients, standard errors, and significance for the mixed effects model in the human and AI condition. Fixed effects are agent's expected value (success rate*desirable outcome - fee), outcome domain (0 = gain, 1 = loss), and trial number.

Mixed-Effects Model of Delegation for Human and AI Agent		
	Dependent variable: delegation	
	Human (1)	AI (2)
Agent's value	2.495*** (0.144)	1.888*** (0.111)
Domain (Loss)	-1.501* (0.610)	0.311 (0.550)
Trial number	-0.034*** (0.010)	-0.032*** (0.007)
Constant	-12.293*** (0.905)	-8.669*** (0.638)
Observations	7860	8190
Log Likelihood	-2138.528	-2529.691
Akaike Inf. Crit.	4293.056	5075.381
Bayesian Inf. Crit.	4348.812	5131.466

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

6. Study 3

Prior research suggests that a perceived loss of control and a fear that agents may act irrationally and opportunistically arises when individuals have private information about their own knowledge while having less precise beliefs and information regarding the agent's actions that lead to the decisions (Lupia, 2015; Owens et al., 2014). Delegation researchers thus recommend reducing the ambiguity and uncertainty about agents' activities so as to increase individuals' trust that agents will not act opportunistically (Spreitzer & Mishra, 1999). This suggests that making a process more transparent and understandable can increase the principal's perceived control (Ananny & Crawford, 2018). However, as the performance of technological systems is generally perceived to be more stable and predictable than human action (Belanche et al., 2020), we expect that increasing transparency about the decision making process will mainly increase delegation rates for human agents, but not for AI agents. In line with this reasoning, recent AI research has shown that high levels of transparency on underlying machine models are indeed unlikely to be of interest or help to users of AI systems (Rai, 2020). Taken together, then, we hypothesize:

H4a. For human agents, process transparency increases delegation rates.

H4b. For AI agents, process transparency does not affect delegation rates.

Study 3 thus aims to confirm that the effect found in the prior studies is due to the social risk and control premiums, which arise from the relative intransparency and unpredictability of human decision processes (Butler & Miller, 2018), and not due to perception of capabilities. In particular, we aim to demonstrate that providing more process information increases delegation rates for human agents but not for AI agents, as the latter do not evoke fears of irrational or maligned actions. Study 3 additionally aims to test our findings in a more realistic setting, namely, when participants are faced with the opportunity to delegate monetarily incentivized investment decisions.

6.1. Design and procedure

Study 3 used a $2 \times 2 \times 2$ (AI vs. human agent; gains vs. losses; high vs. low information transparency) between-subjects experimental design. Participants again had to complete a number of decision tasks. In contrast to prior studies, however, we tested our assumptions in the setting of financial decision making, and participants were told that advisors (both AI and human) operated on behalf of a firm. In this way

we tested whether social risk for AI agents remains low even if the users are informed that the agent acts on behalf of a firm, a fact which might increase perceived intentional uncertainty.

A total of 1100 participants completed the study. Based on the same rules as in the previous studies, 65 participants were removed from the sample (5.9% of the sample), resulting in a final sample of 1054 participants (US MTurkers, $M_{\text{age}} = 39$ years, 40% females). Participants were randomly assigned to a loss or gain condition, to human or AI agents, and to the high or low information condition. The payout rule, expected values of the agents, and the expected value of not delegating followed the same procedure as Studies 1 and 2. In the gain condition participants could earn extra payment for every correct decision. In the loss condition, participants were given extra credits to start with and they lost credits for every wrong decision. In the high information condition, participants were provided with easy-to-understand information that outlined the agent's inner workings and decision logic, an approach we adapted from the medium transparency condition of (Kizilcec, 2016). As in prior studies, participants also received information regarding the agent's accuracy and price. In the low information condition, participants only received information regarding the agent's previous accuracy and the price for delegation. In all conditions, participants were informed that the agents were from the firm "HMS Invest." In this way we tested whether participants would still attribute less intentional capacity to AI agents even after learning that the agent works on behalf of a firm (indeed, the same firm as the human agent). The exact process transparency descriptions for both agents can be found in the Appendix, Fig. 1.

The study was carried out in two parts, as before: a set of learning tasks and a set of delegation tasks (Bobadilla-Suarez et al., 2017). Part I of Study 3 consisted of 6 trials. In each trial, participants received factsheets on two firms and had to choose the firm that was more likely to outperform the other firm in the upcoming period. Following the procedure by Hodge et al. (2021), the factsheets of the firms were multiples of one another, ensuring that no firm was objectively better than the other one (see Appendix, Fig. 2). At the end of Part I, participants were informed of their own success rate in these tasks (50%, as in prior studies). Part II of the study consisted of 20 trials in which participants had to choose whether they preferred to delegate the investment decision to an agent or decide by themselves. This part also included the agent as well as the information manipulations.

After the delegation task and before the participants' learned their final payout, we additionally captured participants' willingness to take risks, to serve as a control in our mixed-model analysis. The exact wording was as follows: "Are you generally a person who is fully prepared to take financial risks or do you try to avoid taking financial risks?" The scale ranged from "1 = not at all willing to take risks" to "7 = very willing to take risks" and was adapted from prior research to fit the financial context (Dohmen et al., 2011). Apart from these differences, the same measures were included in Study 3 as in our prior studies.

6.1.1. Manipulation check

To test whether our manipulation of process transparency was effective, we measured perceived process transparency using the statement "Overall, the investment process steps leading to an advisor's investment decisions are transparent to me" (adapted from Nai et al., 2020, scaled from 1 = completely disagree to 7 = completely agree). Results of an independent *t*-test showed that manipulation was successful and that participants perceived the process to be more transparent in the information condition ($M = 5.61$) than in the no information condition ($M = 4.41$; $t(896.57) = -11.28$, $p < .001$).

6.2. Results

Analysis of delegation rates. We calculated delegation rates as the percentage of trials in which participants chose to delegate. An independent *t*-test with delegation rate as the dependent variable and agent

type as the independent variable showed that, in line with our prior findings, delegation rates for human agents were significantly lower ($M = 0.50$) than delegation rates for AI agents ($M = 0.57$, $t(1029.70) = 4.28$, $p < .001$) (cf. H1). We computed two separate *t*-tests, one for the high information condition and one for the low information condition, again using delegation rate as the dependent variable and agent type as the independent variable. The difference between agents remained significant in the high information condition ($M_{\text{AI}} = 0.58$, $M_{\text{human}} = 0.52$, $t(510.31) = 2.09$, $p = .037$) as well as in the low information condition ($M_{\text{AI}} = 0.57$, $M_{\text{human}} = 0.47$, $t(517.65) = 4.01$, $p < .001$).

To test our key assumptions, we ran a $2 \times 2 \times 2$ ANOVA with agent type (human vs. AI), outcome domain (gain vs. loss) and information transparency (low vs. high information) as independent variables and mean delegation rate as the dependent variable. Results revealed a main effect for agent type ($F(1,1027) = 18.69$, $p < .001$) and outcome domain ($F(1,1027) = 10.74$, $p = .001$), along with significant interaction effects for agent and outcome domain ($F(1,1027) = 6.41$, $p = .011$) and for outcome domain and information transparency ($F(1,1027) = 4.96$, $p = .026$). The main effect for information transparency ($F(1,1027) = 2.72$, $p = .10$), the interaction effect for advisor and information transparency ($F(1,1027) = 1.51$, $p = .220$), and the three-way interaction effect ($F(1,1027) = 0.94$, $p = .333$) were all insignificant. To further analyze the interaction effect for agent and information transparency, we conducted two separate models for the transparency conditions. In the low information transparency model, an independent *t*-test for human agents with delegation rate as the dependent variable and outcome domain as the independent variable showed that delegation rates for human agents were significantly higher in the gain condition ($M = 0.56$) as compared to the loss condition ($M = 0.40$, $t(206.56) = 4.40$, $p < .001$). For AI agents, the same *t*-test showed that the difference was again insignificant ($M_{\text{gain}} = 0.58$, $M_{\text{loss}} = 0.55$, $t(255.20) = 1.01$, $p = .315$). In the high information transparency model, delegation rates for human agents did not differ between the loss and gain conditions ($M_{\text{gain}} = 0.55$, $M_{\text{loss}} = 0.50$, $t(256.95) = 1.247$, $p = .213$). Likewise, the difference was not significant for the AI agent ($M_{\text{gain}} = 0.57$, $M_{\text{loss}} = 0.58$, $t(243.37) = -0.284$, $p = .777$).

In order to further analyze the interaction effect of agent and information transparency, we conducted separate models for the agents. For both models we conducted independent *t*-tests with delegation rate as the dependent variable and information transparency as the dependent variable. Results showed that for human agents, delegation rates were significantly higher in the condition where participants received information regarding the process ($M = 0.52$) as compared to the condition where participants did not receive such information ($M = 0.47$, $t(512.93) = -2.108$, $p = .036$; cf. H4a). For AI agents, this process transparency had no significant effect on delegation rates ($M_{\text{info}} = 0.58$, $M_{\text{no info}} = 0.57$, $t(517.06) = -0.304$, $p = .761$; cf. H4b).

We re-ran our ANOVA model including age and gender as additional determinants. However, the effects of age ($F(1,1025) = 0.20$, $p = .657$) and gender ($F(1,1025) = 0.98$, $p = .323$) were again insignificant and did not change the results of our variables of interest.

Mixed-model analysis of delegation probability. We also modeled the probability of delegation with a mixed-effects model. The model estimates the probability of delegation for an agent's expected value, given the nature of the agent, outcome domain, and information transparency condition. Our outcome variable was the probability that a participant would delegate (delegation = 1), measured at 20 different values of agent's expected value. The estimated coefficients were on a log-odds scale. The model accounts for participants' individual parameter estimates for the agent's expected value (modeled as a random slope) as well as an intercept (also modeled as a random effect, meaning varying base line levels of delegation). We also controlled for trial number to account for linear temporal effects such as disengagement over time, as well as self-stated risk preferences of the participant (Dohmen et al., 2011). Random effects were also included for trial number, but not for risk preferences, given that these are constant across observations for

each participant.

Results of the full model are summarized in Table 6. These show that the agent's expected value positively predicts delegation, meaning that higher expected value leads to an increased likelihood of delegation ($\beta = 1.18$, $SE = 0.02$, $z(20700) = 62.21$, $p < .001$). Furthermore, the model reveals significant interaction effects of agent type and outcome domain ($\beta = -1.19$, $SE = 0.46$, $z(20700) = -2.55$, $p = .011$).

To break down the interaction effect, we ran separate mixed models for AI agents and human agents. We specified the models in the same way as the main model but excluded any main or interaction effect of agent type. Results of these separate models demonstrate that for human agents, the outcome domain had a significant effect on delegation ($\beta = -1.40$, $SE = 0.33$, $z(10300) = -4.18$, $p < .001$). Thus, as proposed, participants in the gain condition were more likely to delegate decisions than participants in the loss condition (cf. H3a). Moreover, there was a significant interaction effect for outcome domain and information transparency ($\beta = 1.01$, $SE = 0.47$, $z(10300) = 2.15$, $p = .031$). That is, as proposed in H4a, participants were more likely to delegate a decision when they received information regarding the agent's investment process (cf. H7a) (see Table 7). For AI agents, only the agent's expected value significantly predicted participants' likelihood to delegate ($\beta = 1.84$, $SE = 0.07$, $z(10400) = 24.56$, $p < .001$). Outcome domain ($\beta = -0.72$, $SE = 0.43$, $z(10400) = -1.65$, $p = .10$; cf. H3b) and information transparency ($\beta = -0.30$, $SE = 0.42$, $z(10400) = -0.72$, $p = .473$; cf. H4b) had no significant effect on delegation to AI (see Table 7). These results further confirm our hypotheses.

6.3. Discussion

The results of Study 3 provide further evidence of the perceptual differences between AI and human agents and their effects on delegation rates. In particular, the results show that, for human agents, the potential decision outcome (gain vs. loss) as well as the level of transparency regarding the agent's decision-making process significantly affect delegation rates. We find that giving participants more information on the decision process increased delegation rates to human agents. This suggests that additional information can indeed help reduce the uncertainties about an agent's actions generally encountered with human delegates. For AI agents, delegation rates were neither domain-dependent nor did they increase when we provided participants with

Table 6

Results of Study 3: Full mixed-effects model of delegation controlling for self-stated risk preferences and trial number. Estimates of the fixed effects, with standard errors and significance. Fixed effects are agent's expected value (success rate*desirable outcome – fee), agent type (0 = AI, 1 = human), outcome domain (0 = gain, 1 = loss), information transparency (0 = low, 1 = high), trial number, and self-stated risk preference.

Full Mixed-Effects Model of Delegation	
	Dependent variable: delegation
Agent's value	1.180*** (0.019)
Agent (Human)	-0.145 (0.334)
Domain (Loss)	-0.303 (0.324)
Transparency (High)	-0.070 (0.311)
Trial number	-0.032** (0.010)
Risk preference	-0.211*** (0.047)
Agent:Domain	-1.185* (0.465)
Agent:Transparency	-0.149 (0.466)
Domain:Transparency	0.348 (0.465)
Agent:Domain:Transparency	0.775 (0.659)
Constant	-3.950*** (0.321)
Observations	20,700
Log Likelihood	-8309.684
Akaike Inf. Crit.	16,645.370
Bayesian Inf. Crit.	16,748.560

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

Table 7

Results of Study 3: Separate mixed-effects models of delegation for AI and human agent. Coefficients, standard errors, and significance for the mixed effects model in the human and AI condition. Fixed effects are agent's expected value (success rate*desirable outcome – fee), outcome domain (0 = gain, 1 = loss), information transparency (0 = low, 1 = high), trial number, and self-stated risk preference.

Mixed-Effects Model of Delegation for Human and AI Agent		
	Dependent variable: delegation	
	Human	AI
Agent's value	1.142*** (0.026)	1.837*** (0.075)
Domain (Loss)	-1.399*** (0.334)	-0.715 (0.435)
Transparency (High)	-0.162 (0.348)	-0.299 (0.418)
Trial number	-0.032** (0.011)	-0.044*** (0.012)
Risk preference	-0.261*** (0.067)	-0.211* (0.092)
Domain: Transparency	1.010* (0.469)	0.809 (0.623)
Constant	-3.723*** (0.409)	-6.614*** (0.589)
Observations	10,300	10,400
Log Likelihood	-4188.742	-3698.062
Akaike Inf. Crit.	8395.484	7418.125
Bayesian Inf. Crit.	8460.643	7497.870

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.

more information on the AI's investment process. Furthermore, the findings of Study 3 are in line with the results of our prior studies, even though we informed participants that both agents acted on behalf of a firm. This suggests that people overly rely on perceptions of their counterpart to infer intentionality rather than consider the broader context in which the respective agent operates (i.e., the intentionality of the firm that might provide decision rules for both human and AI agent).

7. General discussion

7.1. Theoretical implications

The current work makes four novel contributions. First, we have contributed to contemporary research on trust in AI by linking established literature on decision delegation and the social risk and control premiums to the domain of autonomous AI agents. While previous literature has largely focused on participative decision-making and the effects of design, user, or task characteristics on people's trust in AI, our research analyzed users' willingness to entrust decisions with monetary consequences to an autonomous AI and compared this to their willingness to delegate to human agents. In line with the prior notion of algorithm appreciation (Logg et al., 2019), we have shown that users are indeed more willing to cede control to an AI agent as compared to a human agent in 'algorithm-appropriate' domains (i.e., where performance of human and AI agents can be considered equally good). Even more so, individuals in our studies demanded a premium (higher expected value of delegation) in order to cede control to human agents and expose themselves to the attendant social risk (yielding lower delegation rates).

Second, and relatedly, this research revealed the underlying psychological process leading to differences in delegation rates. Following the notion of social agency (Gray et al., 2007), we found that people ascribe less intentional capacity to AI agents than to human agents. Linking this concept with that of social risk (Butler & Miller, 2018), we further showed that these perceptual differences in intentional capacity result in lower perceived social risks when delegating decisions to AI agents. In fact, delegation to AI agents was perceived to be more controllable and to involve less risk than delegation to a social agent, ultimately leading to differences in delegation rates.

Third, we have shown that the strength of the effect of social risk on delegation rates is contextual. Extending prior research on the contextuality of people's trust in AI (Castelo et al., 2019; Goldbach et al.,

2019) and drawing from well-established literature on loss aversion (Kahneman & Tversky, 1979), our research shows that users are less willing to expose themselves to a social risk in the domain of losses as compared to the domain of gains. This difference arises as individuals have a higher desire for control when facing potential losses than when facing potential gains. Given the above-mentioned perceived differences between human and AI agents (i.e. intentionality, risk, and controllability), we find that this higher desire for control in the domain of losses leads to lower delegation rates only for human agents and does not affect delegation to AI.

Finally, this research also contributes to current discussions regarding the explicability of AI. Fueled by Big Data and advancements in data processing, AI in the form of machine learning (especially deep learning) has not only become more powerful but also more of a black box (Huang & Rust, 2021; Rai, 2020). Tremendous research efforts are being exerted to make the intransparent transparent (Eslami et al., 2018). Here, findings of our research show that increased process transparency (i.e., explanations of the inner logic of algorithms) may not only be unfeasible but also undesirable for users. Instead, we found that outcome transparency (i.e., in the form of the success rate of the agent) is more important for fostering trust and thus the willingness to delegate decisions to AI.

7.2. Managerial implications

Our findings offer valuable insights for businesses as to the hows and whys of delegation in new situations where individuals can delegate their decisions to autonomous AI systems. In order to leverage this powerful force for competitive advantage, it will be important for business leaders to understand managers' willingness to entrust AI to make decisions on their behalf (Schrage, 2017). We find that people are willing to forgo potential economic benefits in order to maintain control over decisions in interactions with human agents but not with AI agents. Since the economic environment is not static, it is important to understand not only whether people are more willing to cede control to AI agents, but also how willingness to delegate varies in response to the decision context (i.e., losses vs. gains). Our results show that the difference in delegation rates for human versus AI agents increases when facing potential losses, or—put differently—that AI agents can reduce the under-delegation that is often most pronounced when encountering a loss and when circumstances spur a high desire for control. These findings suggest that managers can introduce AI agents strategically to avoid under-delegation and ease decision making, especially in more aversive situations such as during economic downturns, for risk management, or for projects facing declining economic prospects.

Moreover, our findings suggest that in order to be willing to cede control to a human agent as compared to an AI agent, the human agent must offer either a higher performance accuracy and/or lower fee than the AI agent. Given the same performance accuracy, users' willingness to pay is higher for AI agents, especially in the domain of losses. Considering the higher cost of human labor, these findings are critical for managers. On the other hand, our studies also show that delegation to human agents can be increased by offering operational transparency, hence reducing the social risk and felt uncertainty of delegating to a social agent. Instead of focusing on the explicability of AI, practitioners might do better to consider explaining the decision logic of human agents more clearly so as to reduce perceptions of potentially malign intentions, thus increasing willingness to delegate to agents.

Another related challenge is the amount of procedural transparency users require when interacting with AI. With the increasing advancement of these systems and large amounts of training data, systems not only become more accurate but also less explainable (Chui et al., 2018; Huang & Rust, 2021). The present research demonstrates that while designing for more accurate AI systems (higher expected value of delegation) increases delegation rates, designing for operational transparency does not affect delegation rates. To put it differently, users'

willingness to pay goes up for more accurate AI systems but not for AI systems that simply offer higher process transparency. Rather than communicating the operational process of AI systems, managers should thus focus on clearly conveying their performance and accuracy.

Ultimately then, being capable of leveraging the potential provided by AI is likely to become a key value driver from organizational efficiency and effectiveness. Understanding users' willingness to transfer decision making authority to AI has potentially significant implications across many domains in both private and public life. Designers of AI systems, internal users in an organization, as well as firms and managers can all greatly benefit from understanding when and why people delegate to AI.

7.3. Limitations and future research

Of course, our work comes with limitations which might inspire future work in the following ways: First, our research has focused on the general effect of AI vs. human agents without additionally varying the levels of social cues present. Clearly, this dichotomy does not fully reflect the real world where technology is increasingly humanized and humans are being automatized. Future research could expand our work and examine how an increase or reduction of social cues in both AI or human agents affects the present findings. Our work shows that AI agents (lacking social cues) can reduce the social risk premium demanded for decision delegation. Recently, researchers also found that users respond more favorably to less human-like service robots (Mende et al., 2019) and avatars in video games (Kim et al., 2016), suggesting that inducing social cues and thus social responses may not always be desirable. Taken one step further, we suggest that, instead of asking how to humanize the machine, future research might focus on how (and when) to dehumanize the human. Many human agents already work with computational support systems. Making such procedures more apparent might reduce the gap in delegation rates between AI and human agents and provide a novel perspective for current research on human-machine interactions.

Second, our research deliberately focused on domains and tasks that are 'AI appropriate.' Both the financial and the estimation tasks we used were of a computational nature. Research has shown that people tend to believe that such tasks can be easily and accurately executed through AI systems (Logg et al., 2019) and—especially in banking and finance—many have encountered AI or algorithms before. Other, more subjective domains such as health care could benefit from autonomous diagnosis and medical procedures, but people still feel uncomfortable relying on AI in such contexts (Castelo et al., 2019; Longoni et al., 2019). Given the potentially significant merit of AI in such domains, future research could examine how decision delegation changes in domains where human work is often still considered superior and examine how to design AI systems to foster decision delegation in less algorithmically appropriate domains.

Third, as the main interest of this research was to examine if AI systems can help reduce under delegation, we deliberately provided participants in all studies with precise information about their own and the agent's performance. Clearly, such information is not as readily available in practice or it may not be as easily quantifiable and comparable. However, as an individual's estimation of their own vs. the agent's performance impacts their level of uncertainty and thus their decision to delegate (Lupia, 2015; Owens et al., 2014), future research should examine this aspect further. In particular, it is important to understand how the found effects change, if individuals do not have any prior information about the human and AI agent's performance or are provided with qualitative descriptions about their own and the agent's performance rather than precise performance indicators.

Fourth, researchers could expand our work by examining the economic and societal implications of the effects observed in our studies. For instance, research suggests that people consider human labor more effortful (Buell & Norton, 2011) and less scalable. Despite these notions,

we have found the first evidence that people are not only less likely to delegate decisions to human agents but also less willing to pay for a human agent than AI agents when facing losses. Given the same performance accuracy, participants in our studies were willing to delegate to AI agents at a higher fee (lower expected value of delegation) as compared to human agents. Thus, individuals' willingness to pay, their price sensitivity, and price fairness perceptions might be an interesting focus for future studies on AI in the field. Another pressing concern that leaders and regulators must consider is AI's broader effect on the human, our values, and society as a whole. Increasing delegation to autonomous AI could contribute to depleting human experts from knowledge, experience, and expertise (De Bruyn et al., 2020) and hinder individuality. To date, many recommendations by AI systems are biased towards popularity. An individual's behavior may then be guided by choices of peers, making the popular even more popular and impeding novel content from rising to the top. These filter bubbles can alter how individuals think and act (Pariser, 2011). Developing strategies to allow the benefits of AI advances to be safely realized requires profound understanding of the positive as well as negative consequences on an individual and societal level.

Finally and relatedly, we want to point out that despite the opportunities, delegating complex decisions to AI systems also raises many ethical and legal challenges that cannot be ignored. Ethical concerns include issues of bias, fairness, transparency, and accountability, which

are all inextricably linked to legal issues. Powerful predictive models have the capability to reproduce the biases and prejudices engrained in its training data. Thus, letting these systems function without human control and overriding capability may even lead to more systematic discrimination in the long run (De Bruyn et al., 2020). This does not only have ethical ramifications, but also legal ones, as in the USA and many other countries, discrimination does not need to be intentional to be punishable by law (De Bruyn et al., 2020). Ensuring that the increasing use of autonomous AI systems act as a force for fairness not only requires awareness of these risks, but also methods of limiting them. For this reason, an AI system's transparency, explainability, and interpretability should be a constant and pressing concern for research and practice.

Credit author statement

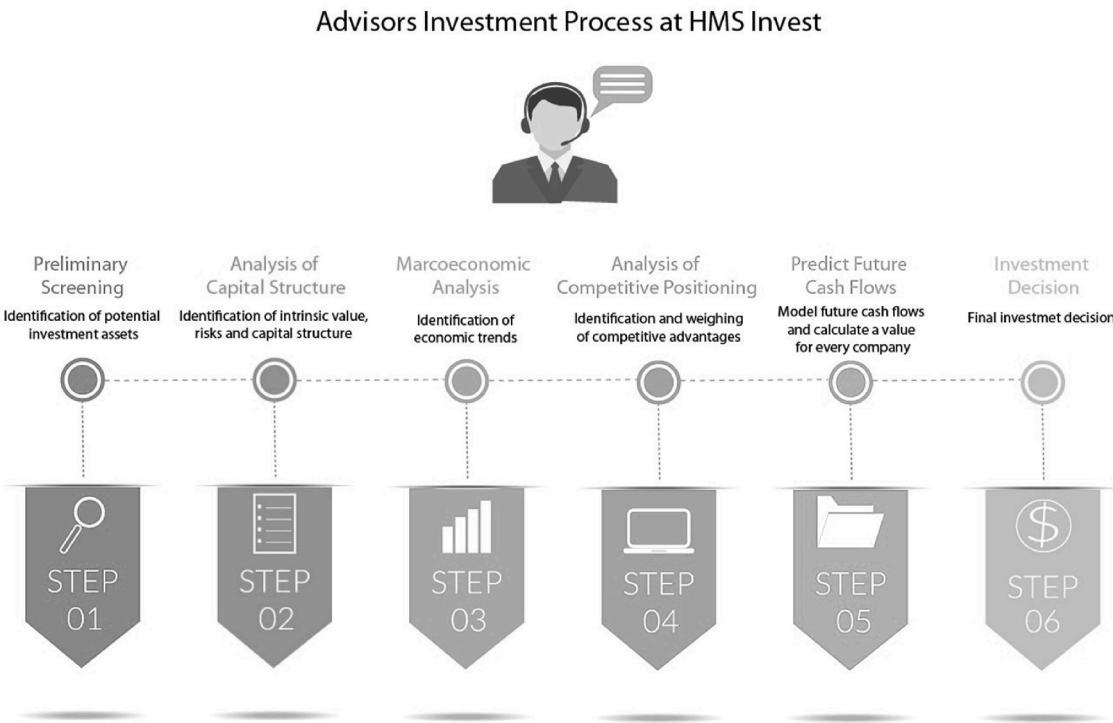
Cindy Candrian: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Anne Scherer: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

None.

Appendix

Study 3: Process Transparency Description in the Human Agent Condition



Study 3: Process Transparency Description in the AI Agent Condition

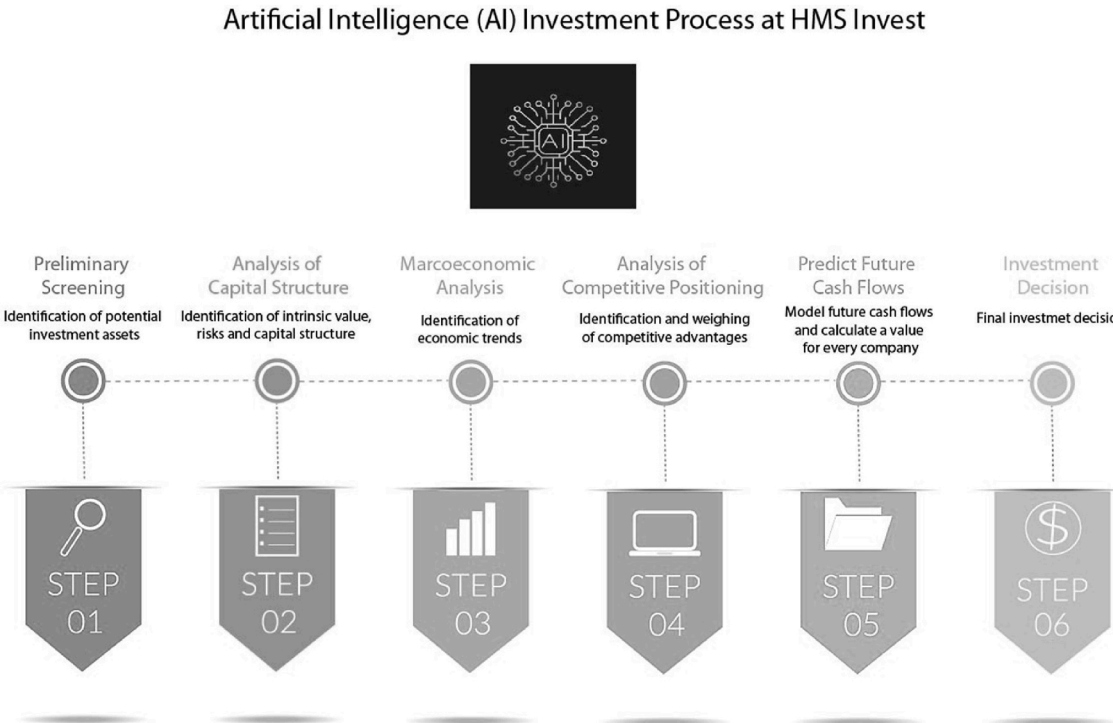


Fig. A1. Explanations used in the high information condition in Study 3 for human (top) and AI (bottom) agent.

Study 3: Firm factsheet shown in trials of Study 3

	Q1	Q2
Revenue	17'986	17'178
Adjusted EBITDA	2'638	2'596
Net Income	1'028	991
Capital Expenditures	901	863
Cash Flow from Operations	1'845	1'789

Fig. A2. Exemplary screenshot of a factsheet participants received. In every trial, they received two factsheets and had to choose the firm that is more likely to outperform the other firm in the upcoming period. Following the procedure by Hodge et al. (2021), the factsheets of the firms were multiples of one another, ensuring that no firm was objectively better than the other.

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